

Senzing

Bootcamp

Completion Recap

A record of everything you built and learned, module by module. Keep it, revisit it, and share it with your team.

BOOTCAMPER Alex Rivera	STARTED 2026-07-16T09:05:00-07:00
COMPLETED 2026-07-16T18:00:00-07:00	PROGRAMMING LANGUAGE Java
PATH Core	PLUGIN VERSION 0.4.0

Modules in this recap

Entity Resolution Concepts	Discover the Business Problem	SDK setup	System verification
Truth Set visualization	Data collection	Data Quality, Mapping, and Transformation	
Data processing	Query, Visualize and Discover		

Run environment

Operating system: macOS 14.5 (Apple Silicon)

Python version: 3.12.3

Language runtime: OpenJDK 21.0.2

Senzing SDK: 4.3.2

Database: SQLite

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Entity Resolution Concepts

Completed July 16, 2026

Information Shared

- What entity resolution is: deciding whether different records refer to the same real-world entity, then matching, relating, and deduplicating them (sourced from Senzing docs via MCP).
- Senzing's principle-based matching (frequency, exclusivity, stability), disclosed vs. discovered relationships, and its differentiators: real-time resolution, no model training, explainability, and scalability.
- The two failure modes (false negatives / false positives) and the conceptual pipeline: standardization -> candidate selection -> scoring -> classification -> entity clustering.

Questions & Responses

- **Q:** Do you have any questions about entity resolution before we continue?
 - **R:** No
- **Q:** Would you like to test your knowledge of entity resolution with a short quiz?
 - **R:** No - skipped the quiz
- **Q:** Are you ready to move on to the next module: Discover the Business Problem?
 - **R:** Yes

Actions Taken

- Presented the Entity Resolution Concepts primer: the concepts banner, an MCP-sourced description of entity resolution, the optional knowledge-check quiz offer, and an open Q&A. This conceptual preamble creates no project files.

End-of-Module Summary

What you accomplished: Introduced the core ideas of entity resolution - principle-based matching, disclosed vs. discovered relationships, and explainability - grounded in Senzing documentation via MCP.

Files produced: (no files - conceptual primer)

Why it matters: Establishes the mental model that every later module (mapping, loading, querying, and the Truth Set visualization) builds on.

Discover the Business Problem

Completed July 16, 2026

Information Shared

- A gallery of common entity-resolution business patterns (Customer 360, Fraud Detection, Data Migration, Compliance/KYC, Marketing, Healthcare, Supply Chain, Vendor MDM, Insurance), backed by real Sensing use-case and case-study documentation.
- How Sensing's principle-based, entity-level, relationship-aware, explainable matching applies specifically to a multi-source compliance/due-diligence scenario.

Questions & Responses

- **Q:** Would you like to see examples of common business problems that entity resolution can solve?
 - **R:** Yes
- **Q:** Do any of these patterns match your situation?
 - **R:** No - requested a generated scenario instead
- **Q:** How would you like to define the business problem?
 - **R:** 3 - I don't have my own data, generate a scenario for me
- **Q:** Does that summary capture your situation accurately?
 - **R:** Yes

Actions Taken

- Generated a bootcamp business scenario (business case offer accepted): Lakeside Community Bank loan-applicant due diligence, backed by real Sensing Las Vegas CORD data (PPP_LOANS, GLEIF, US-LABOR-VIOLATIONS, OPEN-OWNERSHIP; ~100 records each, ~400 total).
- Created docs/business_problem.md (full problem statement).
- Created config/data_sources.yaml (source registry).
- Created docs/data_flow.md (data-flow diagram).
- Created README.md (project overview).
- Auto-created docs/stakeholder_summary_module1.md (one-page executive summary - generated by default, no longer gated behind a question).

End-of-Module Summary

What you accomplished: Defined the bootcamp's business problem - a compliance/KYC due-diligence scenario for loan-applicant vetting - using real Sensing sample data, and documented it for the rest of the bootcamp.

Files produced: docs/business_problem.md, config/data_sources.yaml, docs/data_flow.md, README.md, docs/stakeholder_summary_module1.md

Why it matters: Every later module (SDK setup, mapping, loading, querying) builds directly on this problem statement and these four data sources.

Information Shared

- The Senzing SDK was already installed via Homebrew (version 4.3.2, V4.0+), so installation was skipped; verification and license configuration still ran (both are mandatory).
- Senzing licensing model: the built-in evaluation license allows a limited number of records automatically (confirmed via MCP); a Senzing License Key is needed for larger datasets, installed at licenses/g2.lic.
- macOS Apple Silicon specifics: DYLD_LIBRARY_PATH must be set before JVM launch; the 4.3 cask can miss rpath entries for sqlite/openssl, so those lib paths were added to the env script.
- SQLite is the correct database for bootcamp learning/eval; the DB file and schema are not auto-created and were created explicitly.
- Never modify global shell config; a project-local env script is used instead.

Questions & Responses

- **Q:** Do you already have a Senzing License Key?
 - **R:** No - request a free evaluation license now through the bootcamp
- **Q:** Shall I submit the evaluation-license request with those details (Alex Rivera / Senzing Bootcamp)?
 - **R:** Yes
- **Q:** Has your Senzing License Key email arrived yet?
 - **R:** Yes - installed it
- **Q:** Which database would you like to use?
 - **R:** SQLite

Actions Taken

- Detected the existing Senzing SDK 4.3.2 (Homebrew cask) via szBuildVersion.json; skipped install.
- Created the SQLite database and schema at database/G2C.db (16 tables).
- Created the engine configuration at config/engine_config.json (paths from MCP, DB path overridden to project-local).
- Created the project-local environment scripts src/scripts/senzing-env.sh and src/scripts/senzing-env.bat (SENZING_ROOT, DYLD_LIBRARY_PATH incl. sqlite/openssl, SDK jar, engine config JSON).
- Requested a free evaluation License Key via the MCP submit_feedback tool (delivered by email); installed it to licenses/g2.lic and wired LICENSEFILE into the engine config.
- Wrote and ran src/SetupVerify.java: printed the SDK version (4.3.2) and active license, registered the default Senzing configuration (config id 1375095766), and confirmed the engine connects to the database.
- Created the remaining project directories (licenses/, src/resources/, data/mapping/, data/temp/, build/).
- Recorded license: custom in config/bootcamp_preferences.yaml.

End-of-Module Summary

What you accomplished: Verified the already-installed Senzing SDK works end-to-end in Java - version check,

license check, default-config registration, and database connection - and configured the project's database, engine config, and environment.

Files produced: database/G2C.db, config/engine_config.json, src/scripts/senzing-env.sh, src/scripts/senzing-env.bat, src/SetupVerify.java, licenses/g2.lic

Why it matters: Every later module - loading data in Data processing, querying in Query, Visualize and Discover - depends on a working SDK, a configured database, and the registered default configuration set up here.

System verification

Completed July 16, 2026

Information Shared

- System verification proves the SDK, database, and configuration from SDK setup work end-to-end before real project data enters the pipeline - using a handful of synthetic records designed to resolve deterministically, not the Truth Set (the Truth Set is visualized in its own module next).
- The 8 verification checks and what each confirms: MCP connectivity, SDK initialization, code generation, build/compile, data-source registration, data loading, deterministic results validation, and database operations.
- Register-before-load: the synthetic VERIFY data source code is registered as the default engine config before the load, so the first load never fails with SENZ2207.
- Entity resolution on the synthetic set is known by construction: three records for the same synthetic person (matching name + date of birth + address) resolve to one entity, and a fourth distractor stays its own entity - 4 records -> 2 entities.

Questions & Responses

- {none this module}

Actions Taken

- Generated 4 synthetic verification records into `src/system_verification/verification_data.jsonl` (DATA_SOURCE VERIFY): three for the same synthetic person plus one distractor, designed to resolve into 2 entities.
- Wrote and ran `src/system_verification/verify_init.java` (SDK initialization check - connected to database/G2C.db).
- Generated and built the full pipeline via `generate_scaffold` (`verify_pipeline.java`, compiled with `javac`).
- Wrote and ran `src/system_verification/register_data_sources.java`: registered the VERIFY data source code as the new default config (idempotent) before loading.
- Loaded the 4 synthetic records via `verify_pipeline.java`: 4/4, 0 errors.
- Wrote and ran `src/system_verification/verify_results.java`: confirmed the three matching records resolved to a single entity and the distractor stayed a singleton (2 entities total, exactly as designed); read-by-entity-ID and search-by-attributes both succeeded.
- Ran `src/system_verification/verify_database_ops.java`: write-count, read-by-entity-ID, and search-by-attributes all passed.
- Persisted the 8-check Verification Report to `config/bootcamp_progress.json` (all passed).
- Purged the synthetic VERIFY records via `src/system_verification/purge_verification_data.java`; confirmed zero VERIFY entities remain.

End-of-Module Summary

What you accomplished: Verified the whole Senzing pipeline end-to-end with synthetic records that resolve

deterministically by construction - SDK init, code generation, build, register-before-load, data loading, results validation, and database operations - then purged the synthetic data.

Files produced: src/system_verification/verification_data.jsonl, verify_init.java, verify_pipeline.java, register_data_sources.java, verify_results.java, verify_database_ops.java, purge_verification_data.java

Why it matters: All 8 checks passed - the SDK, database, and configuration from SDK setup are proven to work end-to-end before any real project or Truth Set data is loaded.

Truth Set visualization

Completed July 16, 2026

Information Shared

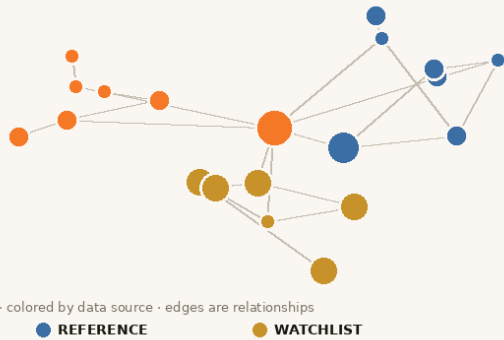
- The Senzing Truth Set (CUSTOMERS 120, REFERENCE 22, WATCHLIST 17 = 159 records) is the deterministic demo dataset with a published ground-truth key (85 expected entities), sourced via the MCP server's `get_sample_data`.
- The interactive visualization is a self-contained web app (six tabs: Entity Graph - with a toggle for just the entities that have relationships - Merge Statistics, Match Keys, Feature Scores, Cross-Source, and Search/Probe) that shows the resolved Truth Set - the "wow moment" of seeing entity resolution work on your own machine.
- Match keys (e.g. +NAME+DOB+PHONE), resolution rules, and relationship types (POSSIBLY_SAME, DISCLOSED, AMBIGUOUS) as surfaced in the resolved-entity JSON via `getEntity(SzRecordKey, SZ_ENTITY_DEFAULT_FLAGS)`.
- On macOS the Python-based bundled app's native SDK is unavailable, so a Java entity export plus an equivalent Python-stdlib/D3 server was used, serving the same four-endpoint contract; it renders fully offline (D3 is vendored in the plugin).

Questions & Responses

- **Q:** Have you finished exploring the visualization?
 - **R:** Yes - done exploring

Actions Taken

- Acquired the 159-record Truth Set (customers.jsonl, reference.jsonl, watchlist.jsonl) via the MCP server.
- Registered the CUSTOMERS/REFERENCE/WATCHLIST data source codes as the default config before loading.
- Loaded all 159 records (0 failures); they resolved into 84 entities with 71 relationships.
- Exported the resolved-entity model (`src/system_verification/ExportEntityModel.java`) and stood up the visualization with the bundled Senzing viz server: standalone snapshot `docs/visualizations/truthset_verification.html` plus a live server, all four API endpoints verified.
- Captured one screenshot per visualization tab and embedded them all in this recap, in the app's tab order.
- Explored the visualization live, then terminated the server (port released).
- Purged the 159 Truth Set records via `SzEngine.deleteRecord(SzRecordKey)`; confirmed zero Truth Set entities remain.



Truth Set entity graph - 159 records resolved into 84 entities across CUSTOMERS, REFERENCE, and WATCHLIST

End-of-Module Summary

What you accomplished: Acquired, loaded, and visualized the Senzing Truth Set - the first interactive "wow moment" - confirming entity resolution works on the workstation, then cleaned up.

Files produced: docs/visualizations/truthset_verification.html, docs/visualizations/truthset_verification-1.png, src/system_verification/ExportEntityModel.java

Why it matters: Seeing the resolved Truth Set - clusters, merges, and cross-source links - makes entity resolution concrete before the bootcamper works with their own data.

Data collection

Completed July 16, 2026

Information Shared

- The bootcamp's data-recommendation hierarchy (CORD first, then free-data repo, then synthesized data) - not needed here since the business problem already chose CORD.
- The license/dataset-size framing: since the bootcamper's Senzing License Key raises the record limit and the collected total (~400) is far below it, no sampling-for-license or SQLite load-time warning applied.
- Each of the four sources uses a genuinely different schema (flat fields vs. nested NAMES/ADDRESSES arrays), setting up the mapping work in Data Quality, Mapping, and Transformation.

Questions & Responses

- {none this module}

Actions Taken

- Downloaded full CORD source files for PPP_LOANS (3,488 records), GLEIF (1,952), US-LABOR-VIOLATIONS (1,554), and OPEN-OWNERSHIP (2,039) from the Senzing-hosted URLs.
- Sampled the first 100 records from each into data/raw/ (ppp_loans.jsonl, gleif.jsonl, us_labor_violations.jsonl, open_ownership.jsonl).
- Validated all four files: readable, non-empty, valid JSONL, correct DATA_SOURCE/RECORD_ID - all passed.
- Updated config/data_sources.yaml to the full tracking schema (file paths, record counts, validation status, provenance, mapping/load status).
- Captured config/cord_metadata.yaml (file hashes/sizes) so Data processing can detect drift before loading.
- Auto-created docs/data_collection_checklist.md (generated by default, no longer gated behind a question) and docs/data_source_locations.md.
- Created .gitignore (excludes data/raw/, data/samples/, database/*.db, licenses/*.lic, build artifacts) and docs/security_compliance.md.

End-of-Module Summary

What you accomplished: Collected the four real CORD data sources planned in the business problem into the project, validated them, and documented their locations and provenance.

Files produced: data/raw/ppp_loans.jsonl, data/raw/gleif.jsonl, data/raw/us_labor_violations.jsonl, data/raw/open_ownership.jsonl, config/data_sources.yaml, config/cord_metadata.yaml, docs/data_collection_checklist.md, docs/data_source_locations.md, .gitignore, docs/security_compliance.md

Why it matters: These four files, each with a different schema, are exactly what Data Quality, Mapping, and Transformation will assess and map onto the Senzing Entity Specification.

Information Shared

- Retrieved the authoritative Sensing Generic Entity Specification (v0.1.0) from the MCP server and saved it to docs/reference/sensing_entity_specification.md as the canonical mapping reference.
- Compared all four sources' fields against the Entity Specification: PPP_LOANS and US-LABOR-VIOLATIONS use flat Sensing attributes (BUSINESS_NAME_ORG/PRIMARY_NAME_ORG, BUSINESS_ADDR_*); GLEIF and OPEN-OWNERSHIP use the nested feature-list structure (NAMES[], ADDRESSES[], IDENTIFIERS[], etc.).
- OPEN-OWNERSHIP is mixed record types (63 ORGANIZATION, 37 PERSON beneficial owners); Sensing reads RECORD_TYPE per record.
- The trade-off of fast-pathing CORD data vs. going through the mapping workflow for hands-on learning: since CORD data is pre-mapped, mapping would re-express an already-correct mapping rather than fix real problems.

Questions & Responses

- **Q:** All four CORD sources are already in Sensing-loadable form. How would you like to proceed?
 - **R:** 1 - fast-path all four to loading

Actions Taken

- Downloaded and saved the Sensing Entity Specification to docs/reference/sensing_entity_specification.md.
- Ran the Sensing-readiness check across 100 sampled records per source: all four passed (valid JSON, DATA_SOURCE + RECORD_ID + name feature present in every record).
- Recorded sensing_ready: true for all four in config/data_sources.yaml; corrected OPEN-OWNERSHIP record_type to MIXED.
- Fast-pathed all four sources: set mapping_status: complete and fast_pathed: true in the registry (no data/sensing-ready/ transformation output was needed, since the sources are already Entity-Specification-compliant).
- Created docs/mapping/data_lineage.yaml with fast-path lineage entries (source_file == output_file, 100 records in == out, 0 rejected, no transformation script).
- Created docs/data_source_evaluation.md documenting the field-level evaluation and categorization of each source.

End-of-Module Summary

What you accomplished: Assessed all four data sources against the Sensing Entity Specification, confirmed they are already Entity-Specification-compliant CORD data, and fast-pathed all four directly to loading (no transformation needed).

Files produced: docs/reference/sensing_entity_specification.md, docs/data_source_evaluation.md, docs/mapping/data_lineage.yaml, updated config/data_sources.yaml

Why it matters: All four sources are validated and confirmed loadable, so Data processing can load them directly and run entity resolution across sources.

Information Shared

- Sensing anti-patterns for loading: initialize the factory/environment once per process; always drain the redo queue using `getRedoRecord()` as the loop sentinel (never `countRedoRecords()`, which does a full table scan); SQLite is dev/test only, not for medium/large production volume.
- Production volume tiers and the license framing: the bootcamper's Sensing License Key covers the bootcamp's volumes, so no capacity concerns regardless of stated production volume.
- The SQLite-at-scale trade-off for a stated medium production tier: proceed on SQLite for the bootcamp now, and migrate to PostgreSQL before real production load (the migration is covered by the graduation production project and migration checklist).
- Why the bootcamp saw 0 PPP_LOANS-to-reference matches: independently-sampled small slices of much larger source files have low odds of specific-company overlap - a sampling-scale artifact, not a pipeline defect - versus the 36 real GLEIF-OPEN-OWNERSHIP matches found (of 37 cross-source matches total; both grounded in the same underlying company registries).

Questions & Responses

- **Q:** In production - not in this bootcamp - how many records do you expect to load?
 - **R:** 3 - more than 500,000 up to 10,000,000, medium tier
- **Q:** Loading may take a while - how would you like to proceed?
 - **R:** 2 - proceed on SQLite for now
- **Q:** Would you like a results dashboard showing entity counts, match statistics, and sample resolved entities?
 - **R:** No

Actions Taken

- Built `src/load/ProductionLoader.java`: a thread-pool loader (medium-tier architecture) with per-record JSON error logging, progress/throughput reporting, and final statistics - adapted from the MCP-sourced `LoadViaFutures.java` pattern.
- First load attempt on PPP_LOANS failed 100/100 with SENZ2207 (data source not registered); diagnosed via `explain_error_code`, fixed by writing and running `src/load/RegisterProjectDataSources.java` to register all four sources.
- Re-ran and loaded PPP_LOANS: 100/100, 0 errors, 228 rec/sec; drained 58 redo records via `src/load/DrainRedoQueue.java`.
- Built `src/load/Orchestrator.java`: a sequential multi-source orchestrator with per-source error isolation, retry with exponential backoff, and health monitoring. Found and fixed an `IdentityHashMap` "Entry was removed" bug (`Map.Entry.getValue()` called after `Iterator.remove()`) during the 10-record sample test, then re-verified before the full run.
- Loaded GLEIF, US-LABOR-VIOLATIONS, and OPEN-OWNERSHIP via the orchestrator: 300/300, 0 errors; drained the coordinated redo queue (0 pending).

- Validated results via `src/query/ValidateResults.java`: 400 records -> 356 distinct entities, 37 cross-source entities; spot-checked 10 with why-matched review (all verified via matching LEI/name/address, no false positives).
- Documented `docs/uat_results.md` and `docs/results_validation.md`, and updated `docs/loading_strategy.md` with final load order, per-source stats, cross-source summary, and issues/resolutions.
- Auto-created `docs/stakeholder_summary_module6.md` (one-page executive summary - generated by default, no longer gated behind a question).
- Updated `config/data_sources.yaml`: all four sources now `load_status: loaded`.

End-of-Module Summary

What you accomplished: Built production-quality single-source and multi-source loading programs, loaded all four data sources into Senzing, processed redo queues, and validated entity-resolution results including real cross-source matches.

Files produced: `src/load/ProductionLoader.java`, `src/load/DrainRedoQueue.java`, `src/load/Orchestrator.java`, `src/load/RegisterProjectDataSources.java`, `src/query/ValidateResults.java`, `docs/uat_results.md`, `docs/results_validation.md`, `docs/stakeholder_summary_module6.md`, updated `docs/loading_strategy.md` and `config/data_sources.yaml`

Why it matters: All 400 records are now resolved in the Senzing database with validated match quality - Query, Visualize and Discover can query, visualize, and explore these results against the original business problem.

Information Shared

- Query requirements were derived directly from the business problem's success criteria (correct resolution without manual cross-checking, why-matched explanations, low false positives) and desired output (per-applicant due-diligence reports).
- Matching-concepts refresher applied to the bootcamper's own data: match keys (e.g. +NAME+ADDRESS+LEI_NUMBER), confidence via SZ_INCLUDE_MATCH_KEY_DETAILS, and real cross-source connections (GLEIF <-> OPEN-OWNERSHIP via shared LEI numbers).
- Quality-evaluation methodology (entity-to-record ratio, possible-match rate, cross-source match rate) computed directly from the exported entity model since no data mart was built at this scale.
- Honest finding: 0 of 100 PPP_LOANS applicants matched a reference source in this run - a sampling-scale artifact (small independent samples of much larger source files), not a pipeline defect - contrasted with 37 real reference-to-reference matches found (36 GLEIF<->OPEN-OWNERSHIP, 1 GLEIF<->US-LABOR-VIOLATIONS: Wynn Las Vegas LLC).

Questions & Responses

- **Q:** Is there anything you'd like to adjust to the derived query requirements?
 - **R:** Add a visualization like the Truth Set one
- **Q:** Would you like a visualization of the resolved entities as an entity graph?
 - **R:** Yes
- **Q:** Would you like to return to Data Quality, Mapping, and Transformation to refine your data mapping?
 - **R:** No - proceed

Actions Taken

- Derived 5 query requirements from docs/business_problem.md success criteria plus the bootcamper's added visualization requirement.
- Built src/query/DueDiligenceReport.java: a per-applicant report using getEntity(SzRecordKey, flags) with SZ_ENTITY_DEFAULT_FLAGS + SZ_INCLUDE_MATCH_KEY_DETAILS; iterates over loaded PPP_LOANS records (never guessed entity IDs). Fixed a flag-type compile error (SZ_ENTITY_INCLUDE_ALL_RELATIONS alone lacks entity names) by switching to SZ_ENTITY_DEFAULT_FLAGS.
- Built src/query/SearchApplicant.java: a search-by-attributes lookup; verified against "Bally Technologies" (found the correct GLEIF entity).
- Ran both programs successfully: 0/100 applicants flagged (explained honestly, not fabricated).
- Computed quality indicators (entity-to-record ratio 0.89, possible-match rate 1.4%, cross-source match rate 10.4%) directly from an exported entity model (data/mapping/project_entities.jsonl, 356 entities) since no data mart exists at this scale; assessed as Acceptable.
- Generated the entity-graph visualization: standalone snapshot docs/visualizations/due_diligence_results.html plus a live server (verified all four API endpoints), captured a

screenshot for the recap, then cleanly stopped it.

- Identified a genuine cross-reference: Wynn Las Vegas LLC matched GLEIF <-> US-LABOR-VIOLATIONS on +NAME+ADDRESS.
- Discover phase (advanced why/how/network tour) explicitly skipped by the bootcamper.

End-of-Module Summary

What you accomplished: Built and ran query programs that directly answer Lakeside Community Bank's due-diligence question, evaluated entity-resolution quality, and produced an interactive entity-graph visualization of the results.

Files produced: src/query/DueDiligenceReport.java, src/query/SearchApplicant.java,
data/mapping/project_entities.jsonl, docs/visualizations/due_diligence_results.html

Why it matters: This is the payoff of the entire bootcamp - the original business problem (manual cross-checking of loan applicants against reference lists) now has working, tested query programs and a visualization built directly on top of validated entity-resolution results.



SENZING BOOTCAMP

Certificate of Completion

Entity Resolution, from first principles to a production pipeline

THIS CERTIFICATE IS PROUDLY PRESENTED TO

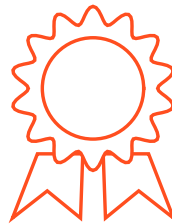
Alex Rivera

for successfully completing 9 modules of the Senzing Bootcamp - Senzing's guided path from defining the business problem to a working entity-resolution pipeline:

Entity Resolution Concepts · Discover the Business Problem · SDK setup · System verification · Truth Set visualization · Data collection · Data Quality, Mapping, and Transformation · Data processing · Query, Visualize and Discover

July 16, 2026

DATE COMPLETED



Senzing Bootcamp

ISSUED BY

Senzing Bootcamp Claude plugin v0.4.0